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THESSALY

Design of FIR Digital Filters:

An Analysis of the Gibbs Phenomenon and the Windowing Method

Jorge Crespo Rivas

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Contents

1	Introduction	2
2	The Ideal Low-Pass Filter	3
2.1	Frequency Domain Specification	3
2.2	Mathematical Derivation of the Impulse Response	3
2.3	Limitations and Realizability Constraints	4
3	Filter Design by Impulse Response Truncation (IRT) and the Gibbs Phenomenon	4
4	The Windowing Method	5
5	The Kaiser Window: An Adjustable Approach	6
6	Conclusion	7
	References	8

1 Introduction

In the modern era of telecommunications, audio processing, and biomedical engineering, Digital Signal Processing (DSP) plays an indispensable role in extracting and shaping information. At the core of these applications lies the digital filter, a mathematical algorithm designed to modify the frequency content of a discrete-time signal. Among the available architectures, Finite Impulse Response (FIR) digital filters have many favourable properties, which is why they are extremely popular in digital signal processing. Unlike their Infinite Impulse Response (IIR) counterparts, FIR filters are inherently stable. More importantly, they can exhibit exactly linear phase, which means that signals in the pass-band will suffer no dispersion. Dispersion occurs when different frequency components of a signal have a different delay through a system, a critical issue in applications where preserving the exact waveform shape is strictly necessary.

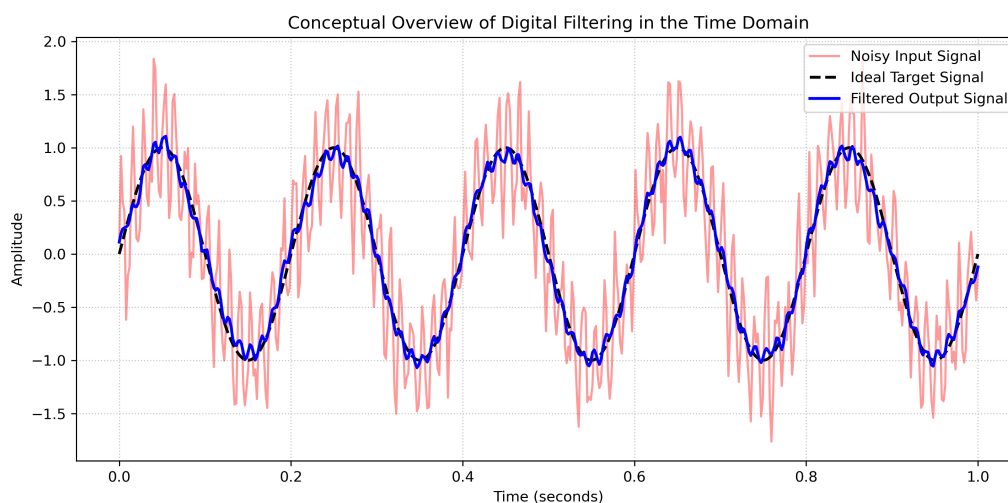


Figure 1: Time-domain digital filtering of a noisy signal.

The design of an FIR filter typically starts by defining the ideal frequency response, $H_D(\Omega)$, such as a "brick-wall" low-pass filter. By applying the inverse Fourier transform, the corresponding ideal impulse response, $h_D[n]$, is obtained, which takes the form of a sinc function. However, this ideal mathematical model presents a fundamental physical limitation: the impulse response theoretically continues forever in both directions and is non-causal. Consequently, it cannot be directly implemented in digital hardware.

The simplest design method to overcome this limitation is impulse response truncation (IRT), which limits the infinite sequence to a finite number of samples. Unfortunately, this abrupt truncation has undesirable frequency-domain characteristics. Truncating the impulse response introduces undesirable ripples and overshoots in the frequency response. This effect is known as the Gibb's phenomenon. Interestingly, as the number of filter coefficients increases, the magnitude of these oscillations does not decrease; they simply become more localized around the discontinuity points. To mitigate these adverse effects, windowing techniques are commonly employed, allowing the designer to smoothly taper the impulse response boundaries to zero.

2 The Ideal Low-Pass Filter

To design a practical Finite Impulse Response (FIR) filter, it is essential to establish a theoretical baseline. This baseline is provided by the ideal low-pass filter, which represents a system that perfectly passes all frequency components below a certain cutoff frequency without any attenuation, while completely eliminating all frequencies above it.

2.1 Frequency Domain Specification

In the discrete-time frequency domain, the ideal low-pass filter frequency response, $H_D(\omega)$, is periodic with a period of 2π . Within the fundamental period $-\pi \leq \omega \leq \pi$, its magnitude response is defined as a perfect rectangular or "brick-wall" function:

$$H_D(\omega) = \begin{cases} 1 & \text{for } |\omega| \leq \omega_c \\ 0 & \text{for } \omega_c < |\omega| \leq \pi \end{cases} \quad (1)$$

where ω_c denotes the normalized angular cutoff frequency, measured in radians per sample.

2.2 Mathematical Derivation of the Impulse Response

The time-domain representation of this system, known as the ideal impulse response $h_D[n]$, can be mathematically derived by applying the Inverse Discrete-Time Fourier Transform (IDTFT) to the frequency specification:

$$h_D[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_D(\omega) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} 1 \cdot e^{j\omega n} d\omega \quad (2)$$

For $n \neq 0$, evaluating this definite integral yields:

$$h_D[n] = \frac{1}{2\pi} \left[\frac{e^{j\omega n}}{jn} \right]_{-\omega_c}^{\omega_c} = \frac{1}{2\pi n j} (e^{j\omega_c n} - e^{-j\omega_c n}) \quad (3)$$

Utilizing Euler's identity, where $\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}$, the expression simplifies directly to the classic sinc relationship:

$$h_D[n] = \frac{\sin(n\omega_c)}{\pi n} = 2f_c \frac{\sin(n\omega_c)}{n\omega_c}, \quad \text{for } n \neq 0 \quad (4)$$

where $f_c = \frac{\omega_c}{2\pi}$ represents the normalized cutoff frequency in cycles per sample.

For the central sample where $n = 0$, direct substitution leads to an indeterminate form (0/0). Applying L'Hôpital's rule or evaluating the initial integral directly for $n = 0$ yields the absolute scale factor:

$$h_D[0] = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} 1 \cdot d\omega = \frac{2\omega_c}{2\pi} = 2f_c \quad (5)$$

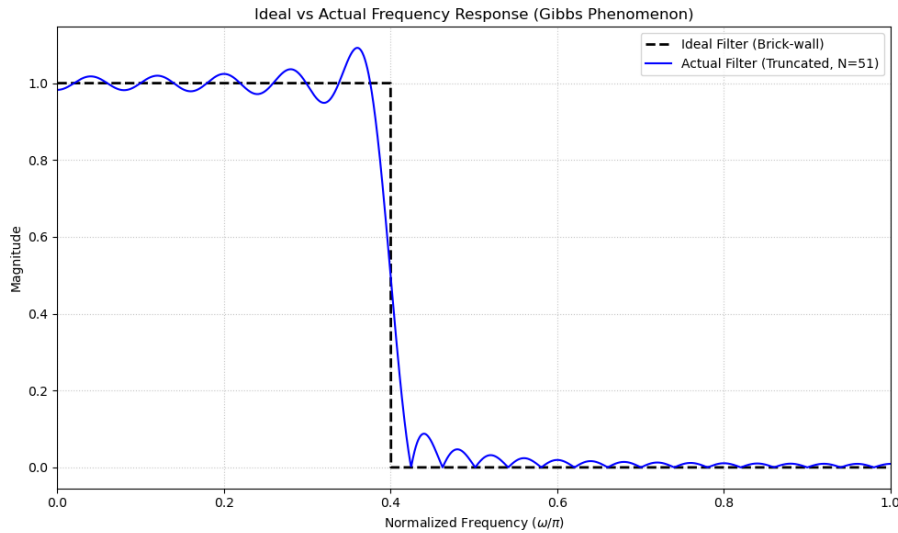


Figure 2: Ideal vs. truncated FIR low-pass filter.

2.3 Limitations and Realizability Constraints

Although the mathematical framework provides a perfect frequency-selective baseline, the resulting impulse response $h_D[n]$ cannot be implemented in any physical hardware system due to two insurmountable constraints:

1. **Infinite Duration:** The sinc function decays at a relatively slow rate of $1/n$ as $n \rightarrow \pm\infty$. Since it never reaches zero, storing or processing the infinite sequence would require unlimited memory and infinite computational operations per sample.
2. **Non-Causality:** The impulse response is symmetric around $n = 0$ and exists for negative time values ($n < 0$). An LTI system is causal if and only if its impulse response is zero for all negative indices. A non-causal system implies that the filter requires knowledge of future inputs to compute the current output sample, which is physically impossible for real-time applications.

3 Filter Design by Impulse Response Truncation (IRT) and the Gibbs Phenomenon

As established, the ideal impulse response $h_D[n]$ is infinite. The most straightforward approach to realizing this filter is Impulse Response Truncation (IRT), which involves retaining only the central $N = 2M + 1$ samples of the ideal response and discarding the rest. Mathematically, this is equivalent to multiplying $h_D[n]$ by a rectangular window.

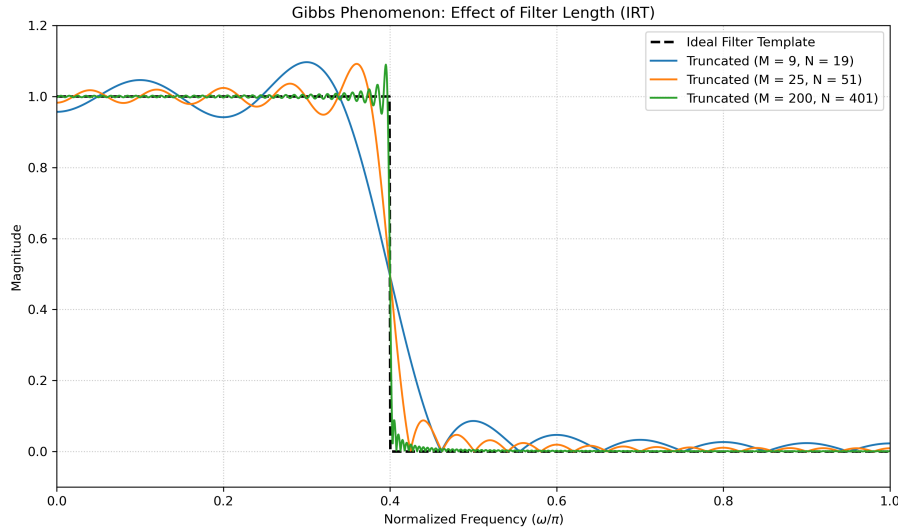


Figure 3: The Gibbs phenomenon for different filter lengths.

While computationally simple, IRT introduces severe distortions in the frequency domain. The abrupt truncation in the time domain causes the frequency response to exhibit oscillatory behavior, known as the Gibbs phenomenon. A critical observation of the Gibbs phenomenon, evaluated across different filter orders in Figure 3, is that increasing the filter order N does not eliminate the amplitude of the ripples. Instead, higher values of N merely increase the frequency of the oscillations and concentrate them closer to the discontinuity point. The maximum overshoot remains nearly constant at approximately 8.95% above the desired passband level. Furthermore, the minimum stop-band attenuation achieved with a rectangular window is limited to approximately 21 dB, which is insufficient for most practical DSP applications where stringent noise rejection is required.

4 The Windowing Method

To suppress the oscillatory behavior caused by the Gibbs phenomenon, the abrupt edges of the rectangular window must be smoothed. This is the foundational principle of the windowing method. By multiplying the ideal impulse response by a tapered window function $w[n]$ that gradually decays to zero at its endpoints, the transition is softened.

In the frequency domain, this time-domain multiplication corresponds to the convolution of the ideal frequency response $H_D(\omega)$ with the Fourier transform of the window function $W(\omega)$. The spectral characteristics of $W(\omega)$ —specifically its main-lobe width and its side-lobe amplitude—directly dictate the final performance of the FIR filter. As observed in the spectral plot, there is an inescapable design trade-off where a narrower main-lobe yields a sharper cut-off, while lower side-lobes reduce the Gibbs ripples at the cost of a wider transition bandwidth.

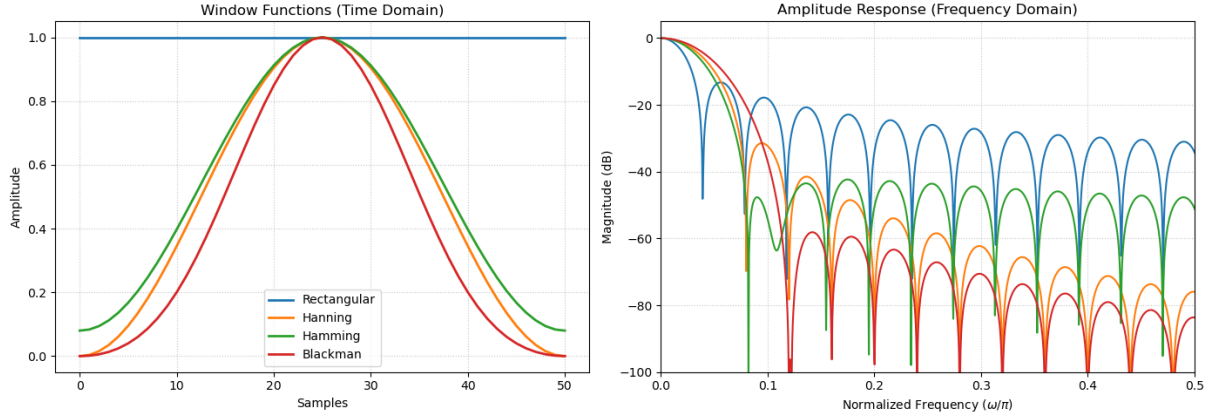


Figure 4: Standard window functions in time and frequency domains.

For instance, while the Rectangular window has the narrowest main-lobe, its side-lobe attenuation is extremely poor (-13 dB). Conversely, the Blackman window provides excellent side-lobe attenuation (-57 dB), ensuring minimal ripple, but forces a much wider transition band due to its broad main-lobe. Because these standard windows have fixed characteristics, a designer cannot independently control the transition width and the ripple amplitude without switching to a different window family entirely.

5 The Kaiser Window: An Adjustable Approach

To overcome the rigid limitations of standard window functions, James Kaiser developed an adjustable window based on the zeroth-order modified Bessel function of the first kind (I_0). The mathematical definition of the Kaiser window is expressed as:

$$w[n] = \frac{I_0 \left(\beta \sqrt{1 - (2n/(N-1))^2} \right)}{I_0(\beta)}, \quad \text{for } -\frac{N-1}{2} \leq n \leq \frac{N-1}{2} \quad (6)$$

where N represents the window length and β is a continuous shape parameter that governs the trade-off between the main-lobe width and the side-lobe attenuation level.

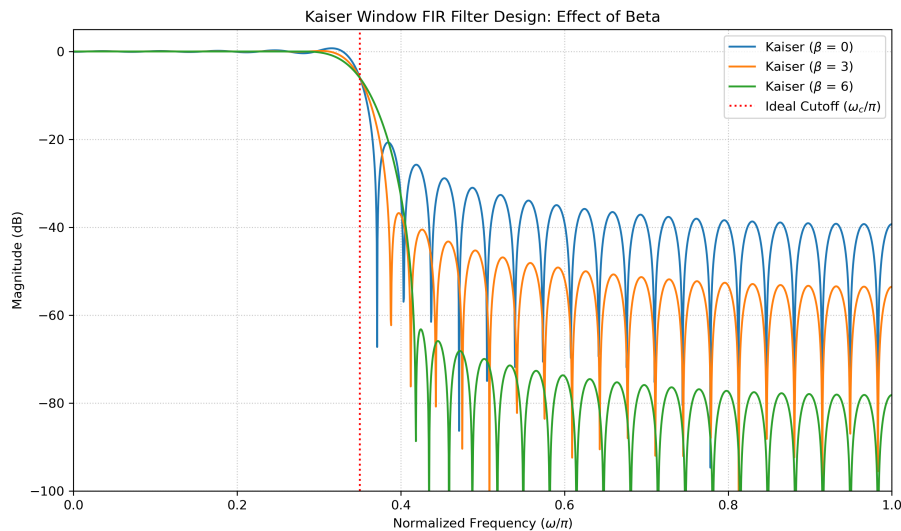


Figure 5: Effect of the continuous shape parameter β on a Kaiser window filter.

The primary advantage of the Kaiser window lies in its empirical flexibility. By adjusting β , the designer can seamlessly tune the performance of the filter to meet precise specifications. As simulated in Figure 5, when $\beta = 0$, the window behaves exactly like a classic Rectangular window, offering the narrowest transition band but the highest side-lobe ripples. As β increases to 3 and 6, the window function becomes heavily tapered in the time domain. Consequently, the side-lobes in the frequency domain are drastically suppressed, drastically increasing the minimum stop-band attenuation at the expense of a wider, more gradual transition band. This continuous control enables a precise optimization that standard fixed windows cannot achieve.

6 Conclusion

Throughout this work, the structural and mathematical properties of Finite Impulse Response (FIR) digital filters have been systematically evaluated. FIR architectures provide significant advantages in digital signal processing, particularly due to their guaranteed stability and exactly linear phase behavior, which completely eliminates phase dispersion across the passband.

However, translating an ideal low-pass brick-wall response into a physically realizable system introduces distinct design challenges. Direct truncation of the infinite sinc sequence via the IRT method triggers the Gibbs phenomenon, causing persistent 8.95% overshoots that cannot be attenuated by simply increasing the filter order. While standard window functions smoothly suppress these oscillations, they introduce a rigid trade-off between cut-off sharpness and ripple rejection.

Ultimately, the Kaiser window provides the most robust solution for practical FIR filter design. By introducing an adjustable shape parameter β , it removes the limitations of fixed windows, granting designers the mathematical flexibility to independently tune and balance the transition bandwidth and stop-band attenuation based on specific hardware constraints.

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